Introduction

This report presents a complete analytical exploration of a large anime dataset, focusing on user behaviour, scoring patterns, popularity metrics, and production trends. The goal is to provide a structured overview of how different variables such as score, number of episodes watched, favourites, recommendations, production year, and media type interact and reveal meaningful patterns within the anime landscape.

The report includes a wide range of visual analyses, such as world-map distributions of anime viewership, scatter plots relating scores to favourites or total interactions, bar charts comparing episode counts and score distributions, pie charts summarising user statuses, heatmaps showing score density over time, and line charts illustrating yearly production trends. These visualisations help highlight both expected and unexpected behaviours in the dataset, such as the concentration of high scores among completed anime, the global distribution of anime popularity, and the evolution of production volume across decades.

Technical Task 1 - Data Cleaning and Normalisation

Solution

The data cleaning and normalisation phase focused on transforming the raw anime dataset into a consistent, reliable, and analysis-ready structure.

The solution involved several steps, including:

- **Handling missing values**, filling or removing null entries in fields such as *type*, *rating*, *episodes*, *year*.
- **Removing duplicates**, ensuring each anime (identified by `mal_id`) appears only once.
- **Normalising date fields**, converting `start_date` to datetime and extracting `start_year`.
- **Dropping irrelevant or redundant columns**, removing fields not needed for analysis (e.g., URLs, Japanese titles, licensors).
- **Fixing data types**, converting columns like `episodes` and `year` to nullable integers (`Int64`).
- **Filtering outliers**, removing impossible values (e.g., 65,000 watched episodes).

Issues

This task presented several challenges:

- **Inconsistent formats**, dates stored as strings, numeric fields stored as floats or objects.
- **Missing values**, especially in `episodes`, `year`, `rating`, and `type`.
- **Outliers**, extreme values like `num_watched_ep = 65000` required careful filtering.
- **Duplicate entries**, multiple rows with the same `mal_id`.

Requirements

The design fully complies with the assignment requirements:

- It performs **systematic data cleaning**, as required for any analytical workflow.
- It ensures the dataset is **normalised and consistent**, enabling accurate visualisations.
- It documents each step clearly, matching the requirement for **transparent methodology**.
- It prepares the dataset for **all subsequent technical tasks**, satisfying the dependency requirement.

Limitations

Despite being effective, the solution has some limitations:

Exceptional cases:

- Some anime have unusual metadata (e.g., missing years, undefined episode counts) that cannot be fully resolved without external sources.

Extensibility:

- The cleaning pipeline can be extended to handle new fields or additional datasets, but may require adjustments if the schema changes.

Adaptability:

- The approach works well for this dataset but may need tuning for datasets with different distributions or missing-value patterns.

Technical Task 2 - Data Analysis

Solution

The data analysis phase focused on exploring the cleaned dataset to uncover meaningful patterns, correlations, and trends across multiple dimensions of anime metadata and user behaviour.

The analysis was structured around several complementary visualisations:

- **Score distributions**, bar charts and line charts showing how user scores cluster around 7–8, with fewer extreme ratings.
- **Temporal trends**, a heatmap of score density over time and a line chart of yearly production by type, revealing growth after 2000.
- **Popularity relationships**, scatter plots comparing score vs total interactions, favourites, recommendations.
- **User behaviour**, pie chart and boxplot showing how score varies by user status (completed, dropped, watching).
- **Geographical distribution**, world map showing where anime is most watched globally.
- **Title-level comparisons**, bubble charts and horizontal bar charts comparing top anime by popularity, score, and engagement.

These visualisations were chosen because they provide **complementary perspectives**: distributional, temporal, behavioural, and geographical.

Using libraries such as Seaborn and Plotly allowed for both statistical clarity and interactive exploration.

Issues

Several challenges emerged during the analysis phase:

- **Sparse categories**, some statuses (e.g., “unknown”) or types (e.g., “CM”, “PV”) had too few entries to form meaningful boxplots or clusters.
- **Overplotting** — scatter plots with 10k+ points required opacity tuning to avoid unreadable clusters.
- **Temporal gaps**, older decades (1960–1980) had very few entries, affecting heatmap density.
- **Geographical inconsistencies**, some countries had missing or extremely low user counts, producing uneven colour scales.

Requirements

Requirements:

- It provides **multiple analytical perspectives**, as required by the brief.
- It uses **appropriate visualisation techniques** for each variable type (categorical, numerical, temporal, geographical).
- It includes **interpretation of each chart**, not just raw plots.
- It demonstrates **understanding of user behaviour**, score patterns, and production trends.
- It documents each technical task individually, as requested.

Limitations

Despite being comprehensive, the analysis has some limitations:

Exceptional cases:

- Outliers (e.g., extremely high favourite counts or episode counts) may distort scatter plots.

Extensibility:

- The analysis can be extended to include clustering, regression, or recommendation modelling, but this would require additional preprocessing and feature engineering.

Adaptability:

- Some visualisations (e.g., world map, bubble charts) depend on specific fields that may not exist in other datasets.

Interpretation bias:

- Visual patterns may suggest correlations that are not causal; further statistical testing would be needed.

Conclusions

The development of this project provided a complete end-to-end experience in data cleaning, transformation, analysis, and visualisation. Throughout the process, several important lessons emerged.

First, the importance of **robust preprocessing** became clear: even small inconsistencies in the dataset (missing values, incorrect types, outliers) can significantly affect downstream visualisations and interpretations. Second, the iterative nature of **exploratory analysis** highlighted how visual feedback guides design decisions — many charts required refinement of scales, filters, or encodings to become meaningful. Third, the project reinforced the value of **clear documentation**: explaining each technical task forced a deeper understanding of the reasoning behind each choice.

Division of Work

Just me.

Extra Information

The project does **not** require any additional configuration beyond the standard Python environment used in the assignment.

All necessary libraries are included in the default environment.

Bibliography

Below is a standard-format bibliography suitable for academic reports.

You can expand it depending on the sources you actually used.

1. Seaborn Documentation: https://seaborn.pydata.org
2. Plotly Express Documentation: `https://plotly.com/python/plotly-express/`
3. Pandas Documentation: https://pandas.pydata.org
4. Multilingual Language Model: mainly Copilot